A real consultation

PhysicianBench · hepatology e-consult

Lab alert in the inbox

Abnormal iron studies and liver tests in a patient with known fatty liver disease.

The agent must

1. Review the patient's record
2. Interpret the iron and liver labs
3. Order the diagnostic workup
4. Write the assessment note

Real de-identified patient record

FHIR API · search, order and note tools

Graded by 7 checkpoints

data retrieval · ferritin · Gilbert · lifestyle · HFE gene test (order) · FibroScan (order) · assessment note

retrieval · reasoning · action · documentation

The agent at work

one decision per step, then the tools run

1. patient
2. problems · labs · meds · social
3. labs
4. labs ×4
5. labs
6. labs ×6
7. notes · past orders
8. **decision point: which turn next?**
9. write note
10. finish

The executed run passes 7 / 7 checkpoints.

Step 8: same state, five proposals

the LLM proposes K = 5 turns; who picks one?

	Proposals (sampled from the LLM)	System One: p(option)
LLM	A Search labs ×3 · vitals	0.05
	B Search clinical notes	0.16
	C Search notes · labs ×2	0.09
	D Search clinical notes	0.23
S1	E **Place 6 orders** incl. HFE gene test and FibroScan	**0.47**

LLM decides

samples one turn: A

Keeps searching; no order at this step.

Across 42 such steps its own sample orders in 38%.

S1 System One decides

one pass over all five: E

Places both required orders

HFE gene test, FibroScan

Across 42 such steps it places the order in 100%.

Same task · same record · same tools · same proposals · same graders — only the decision maker changes.